

Mobile Systems & Edge Intelligence

Phones, wearables, IoT devices, vehicles—and
beyond

Smartphones are our primary lens

Yaxiong Xie

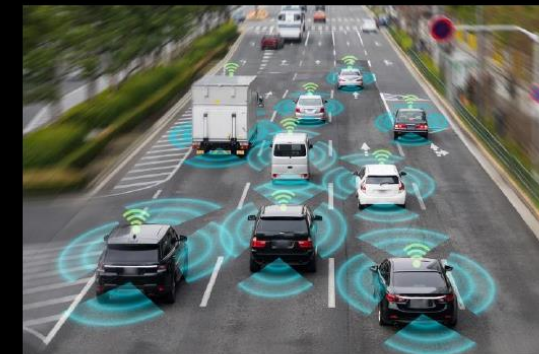
Fall 2026

Wireless biomedical devices

WIRELESS IMPLANTABLE MEDICAL DEVICES



Wireless vehicles



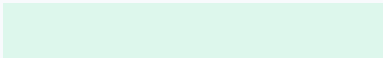
Mobile systems operate in changing contexts

01



Phone

Connected and
sensor-rich



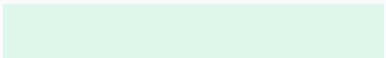
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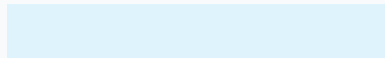


02



Wearable

Moves with
the user



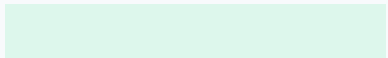
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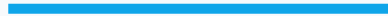


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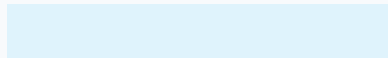


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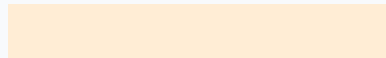


03



Embedded

IoT devices at
the edge



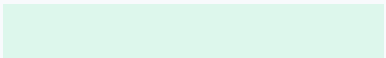
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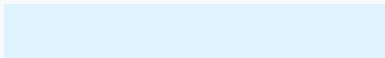


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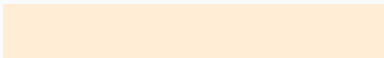


03



Embedded

IoT devices at
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04



Medical

Monitors and
responds



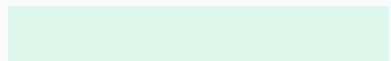
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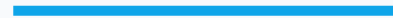


Phone

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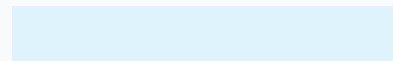


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Wearable

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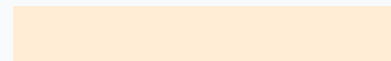


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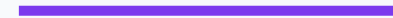


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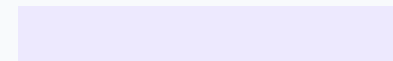


04



Medical

Monitors and
responds



05



Vehicles

Robots move
through
the world



**“Mobile” describes the operating context—not
just the form factor.**

Four capabilities power mobile applications

01

Communication

Hardware: cellular ·
Wi-Fi · Bluetooth

Apps: maps · calls ·
updates

Four capabilities power mobile applications

01

Communication

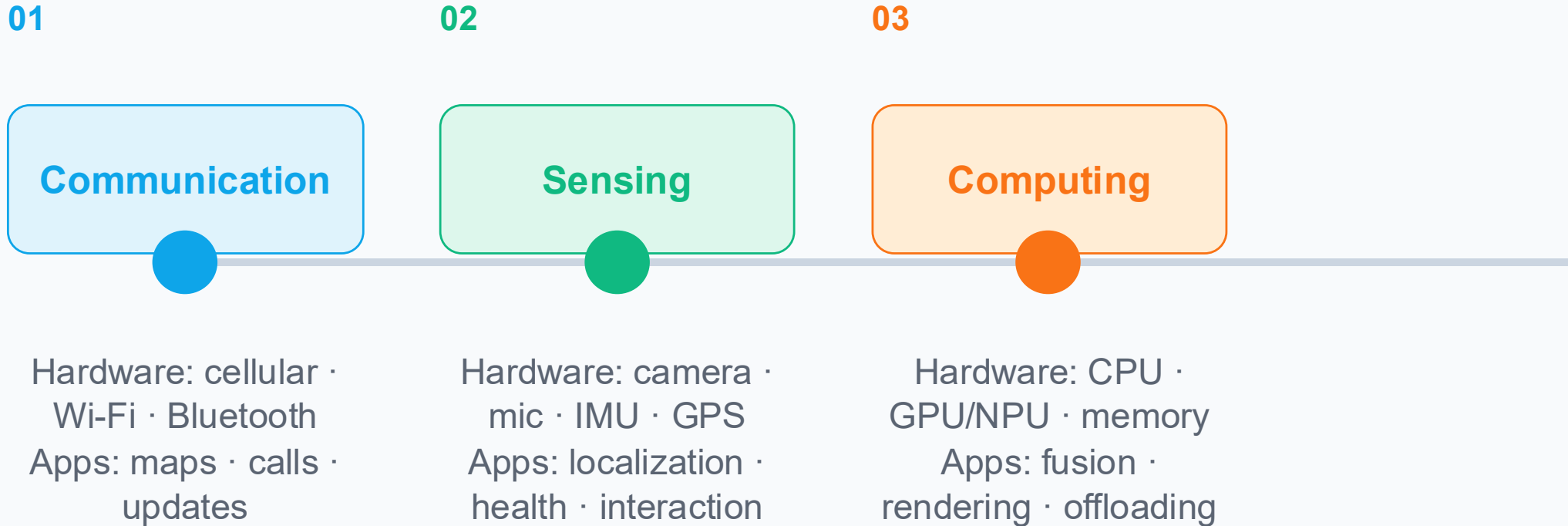
Hardware: cellular ·
Wi-Fi · Bluetooth
Apps: maps · calls ·
updates

02

Sensing

Hardware: camera ·
mic · IMU · GPS
Apps: localization ·
health · interaction

Four capabilities power mobile applications



Four capabilities power mobile applications

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Communication

Hardware: cellular ·
Wi-Fi · Bluetooth
Apps: maps · calls ·
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02

Sensing

Hardware: camera ·
mic · IMU · GPS
Apps: localization ·
health · interaction

03

Computing

Hardware: CPU ·
GPU/NPU · memory
Apps: fusion ·
rendering · offloading

04

Intelligence

Models: recognition ·
prediction · planning
Apps: assistants ·
navigation ·
adaptation

A useful mobile system combines all four—not one in isolation.

Walking navigation needs the full mobile system



A familiar navigation app
already needs all four:

Communication Live maps + traffic

Walking navigation needs the full mobile system



A familiar navigation app already needs all four:

Communication

Live maps + traffic

Sensing

GPS · camera · IMU

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A familiar navigation app already needs all four:

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Live maps + traffic

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GPS · camera · IMU

Computing

Fuse streams + localize

Walking navigation needs the full mobile system



A familiar navigation app already needs all four:

Communication

Live maps + traffic

Sensing

GPS · camera · IMU

Computing

Fuse streams + localize

Intelligence

Choose route + guide

Walking navigation needs the full mobile system



A familiar navigation app already needs all four:

Communication

Live maps + traffic

Sensing

GPS · camera · IMU

Computing

Fuse streams + localize

Intelligence

Choose route + guide

The experience works only when all four cooperate.

Why is navigation a mobile systems problem?

What keeps changing?

User location

Wireless conditions

Sensor observations

Battery + workload

What must the app still deliver?

- Correct guidance
- Fast response
- Privacy
- Energy efficiency

Phones are our primary lens; the same principles apply to wearables, IoT devices, and vehicles.

A mobile system closes a loop under changing conditions

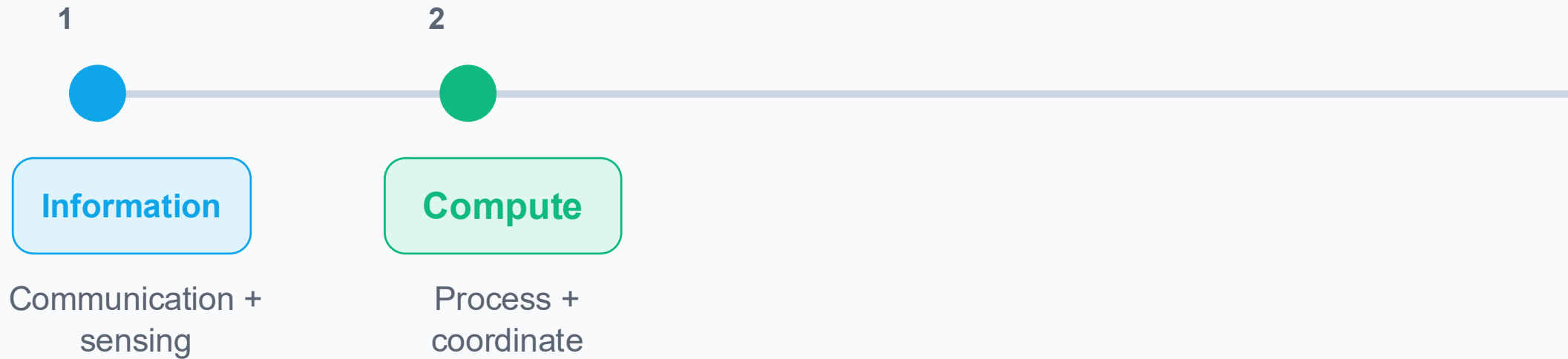
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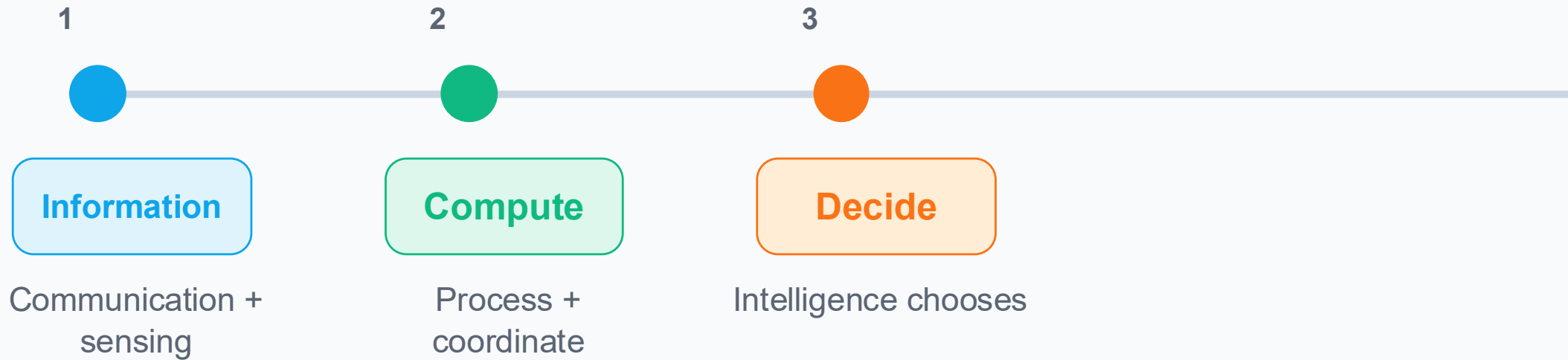
Information

Communication +
sensing

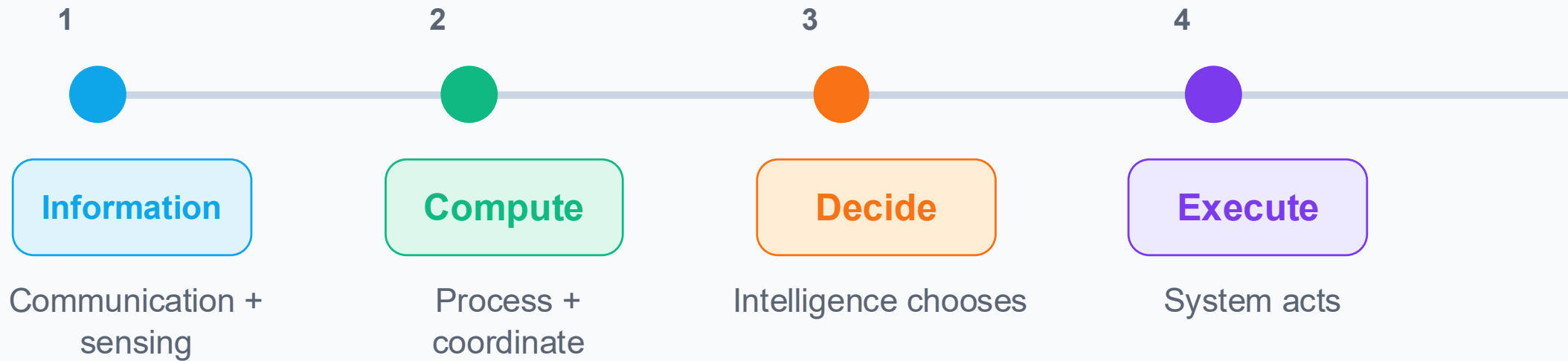
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A mobile system closes a loop under changing conditions

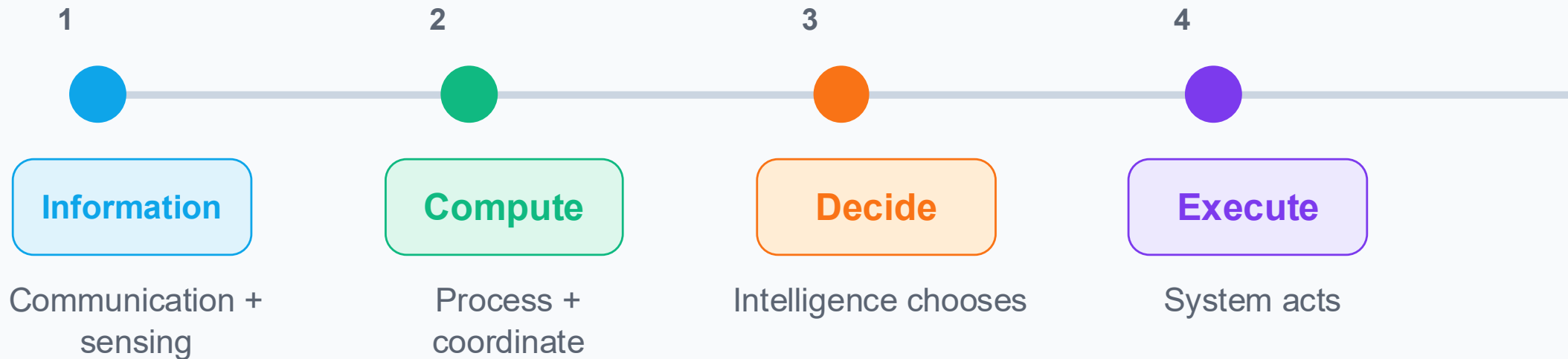


A mobile system closes a loop under changing conditions



A mobile system closes a loop under changing conditions

FEEDBACK ← The environment changes; the next cycle begins.



01

Communication moves information

Mobile devices exchange data with nearby devices, infrastructure, edge servers, and the cloud.

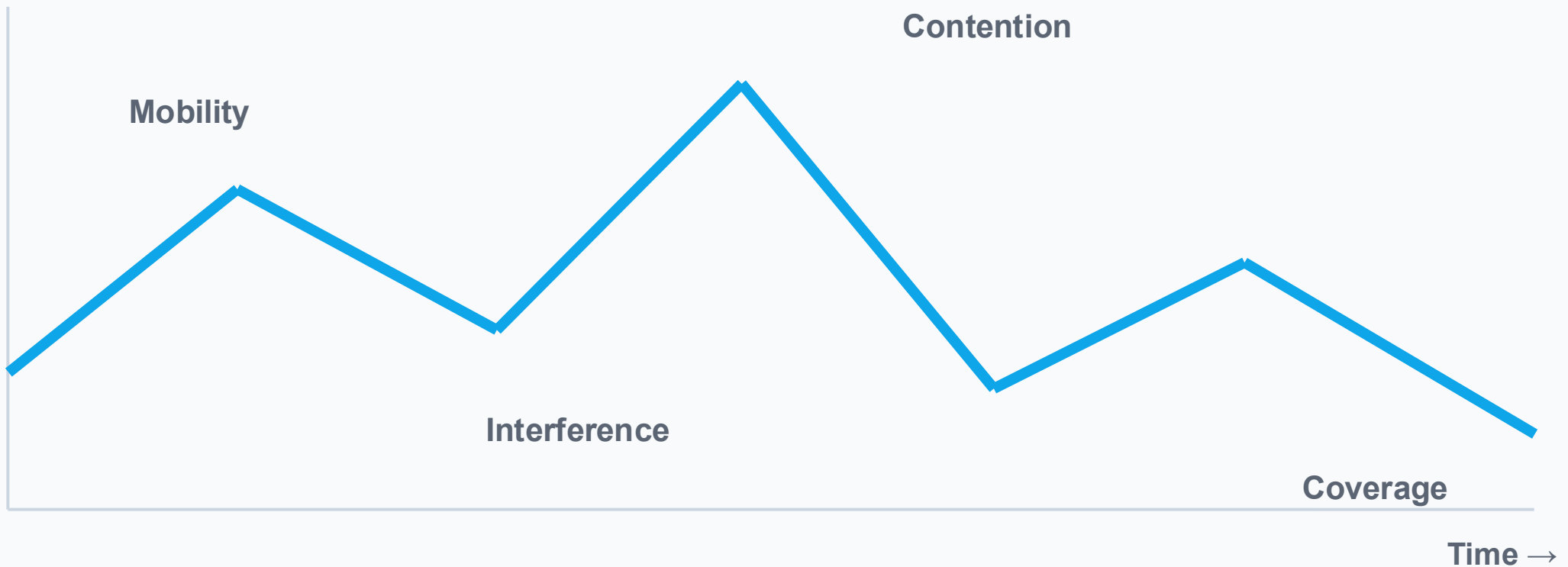
Wireless links

Changing channels

Adaptive applications

Wireless changes while the app is running

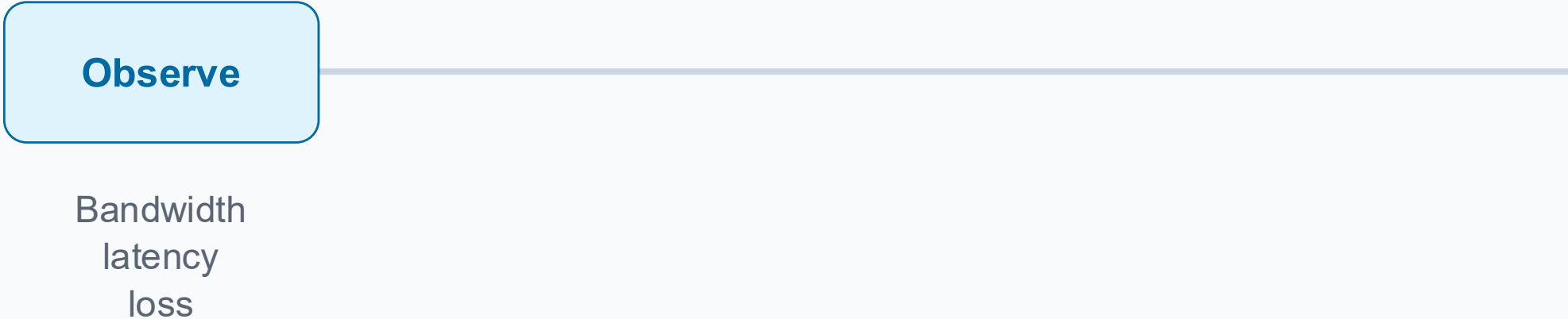
Available capacity



A mobile application must operate across the entire curve—not only at the average.

Mobile apps must adapt—not just transmit

Observe



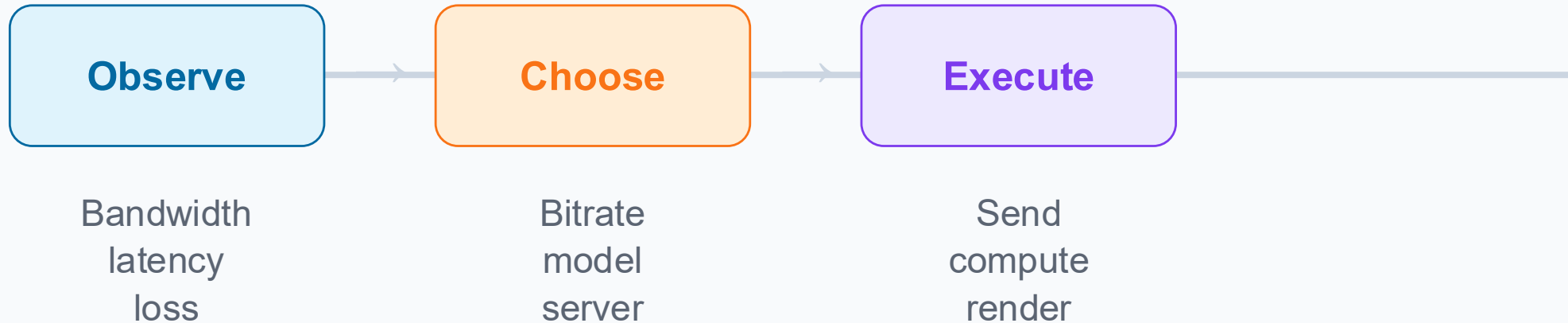
```
graph LR; A[Observe] --- B[ ];
```

Bandwidth
latency
loss

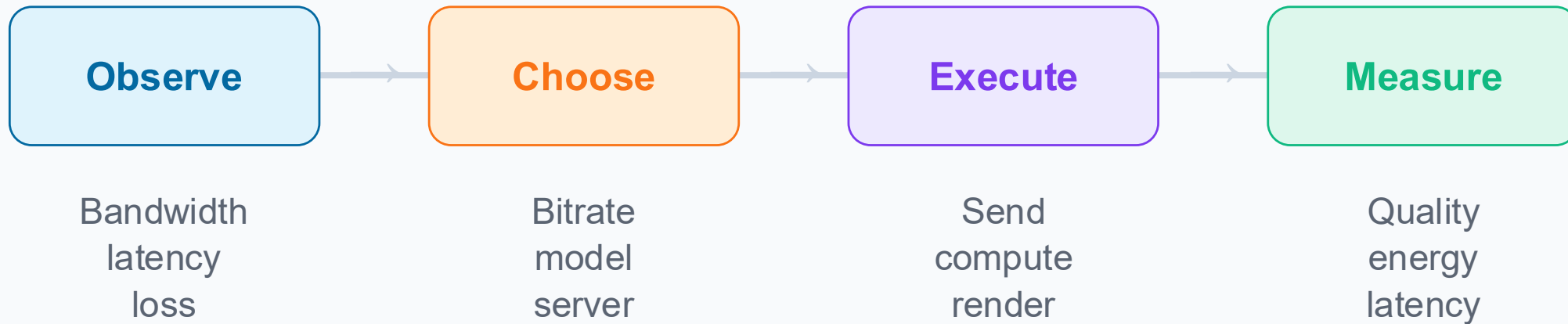
Mobile apps must adapt—not just transmit



Mobile apps must adapt—not just transmit



Mobile apps must adapt—not just transmit



↻ The result updates the next decision

02

Sensing turns signals into context

Cameras, microphones, IMUs, location, and radio signals reveal the physical world around a device.

Sensor modalities

Signal pipeline

Wireless sensing

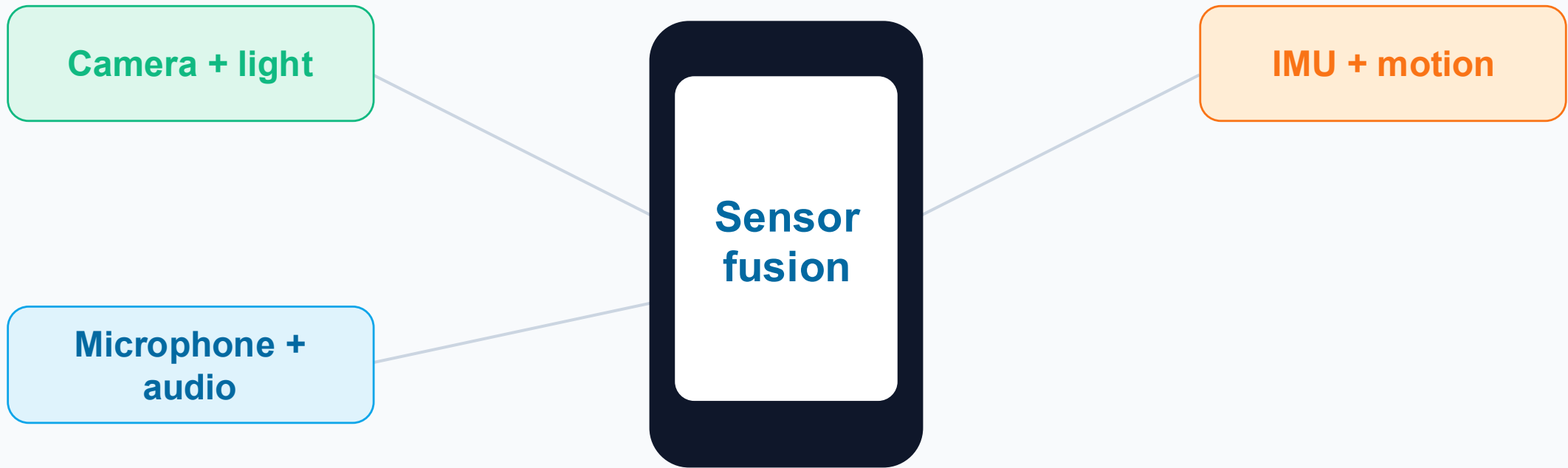
A phone carries a multi-modal sensor suite



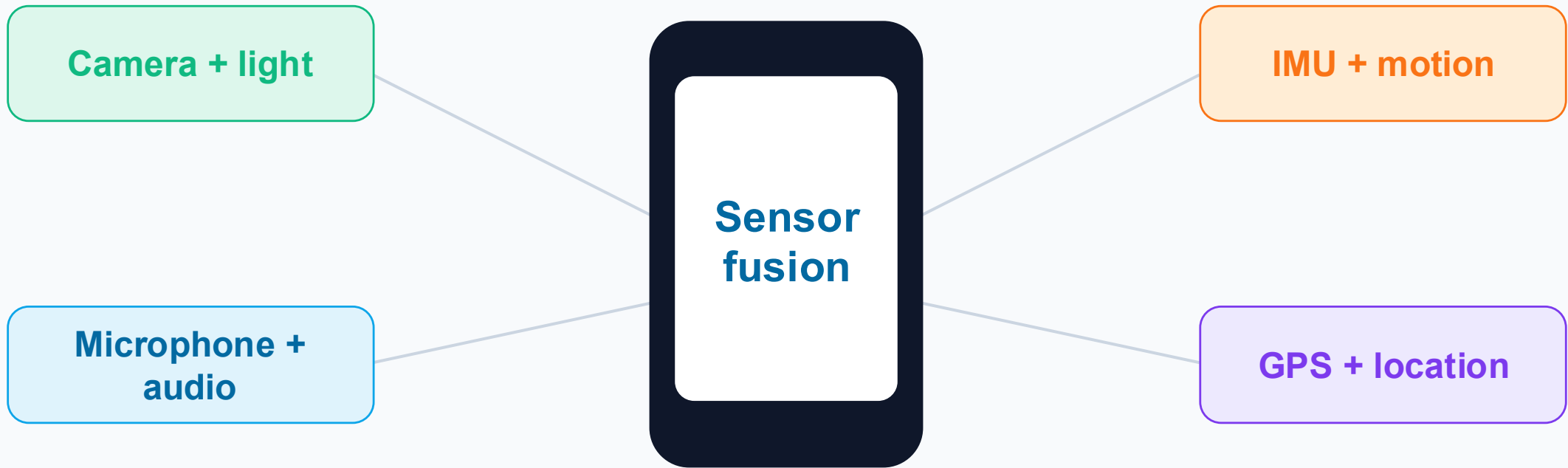
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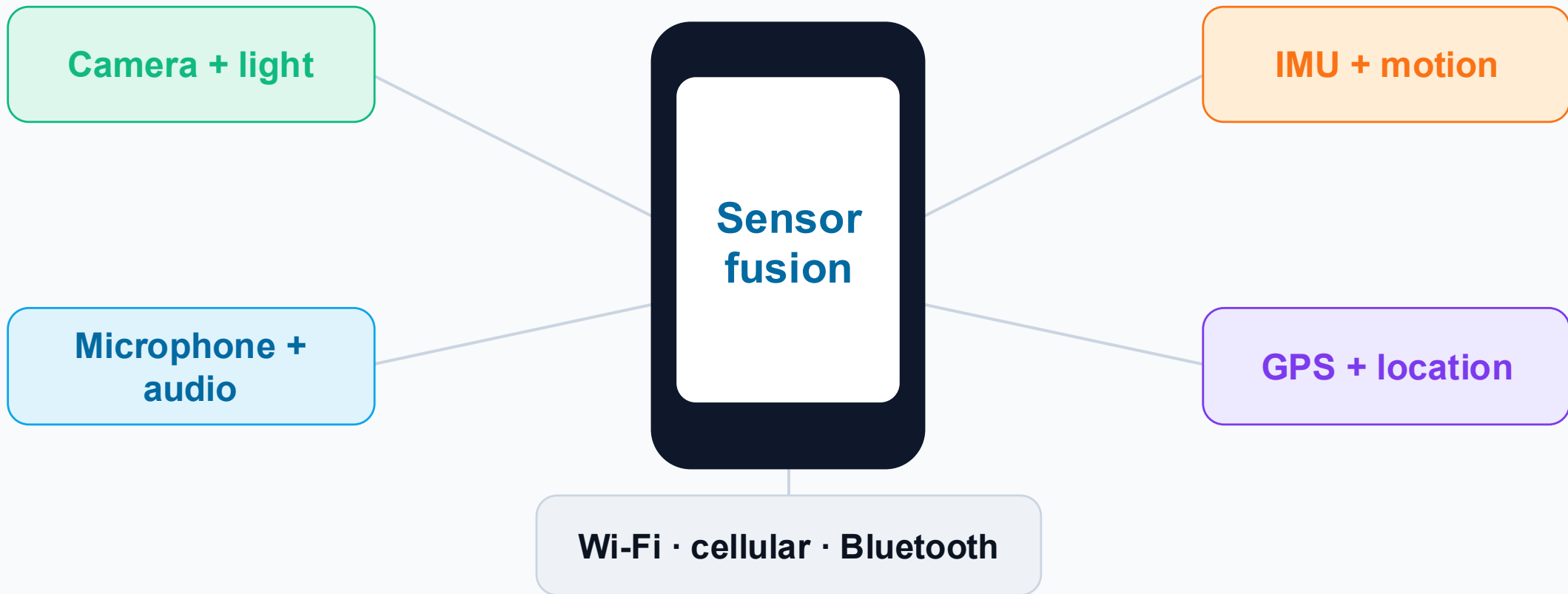
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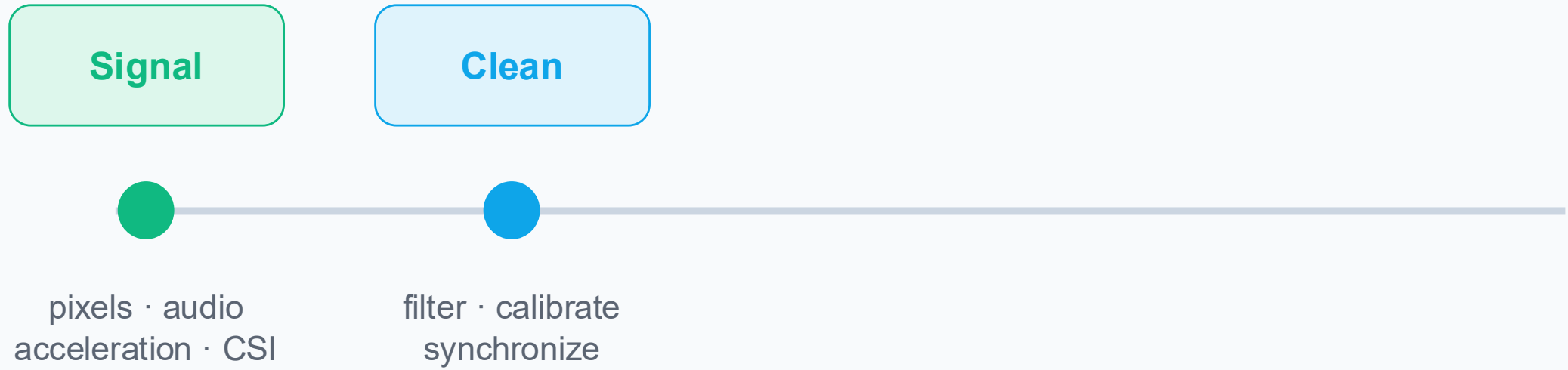
Raw signals require interpretation

Signal

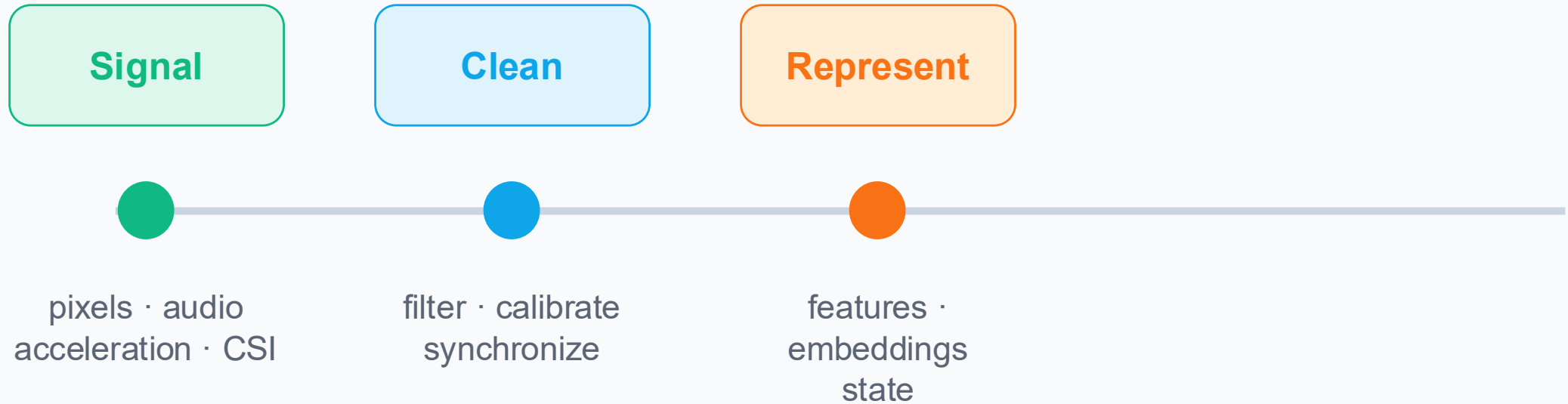


pixels · audio
acceleration · CSI

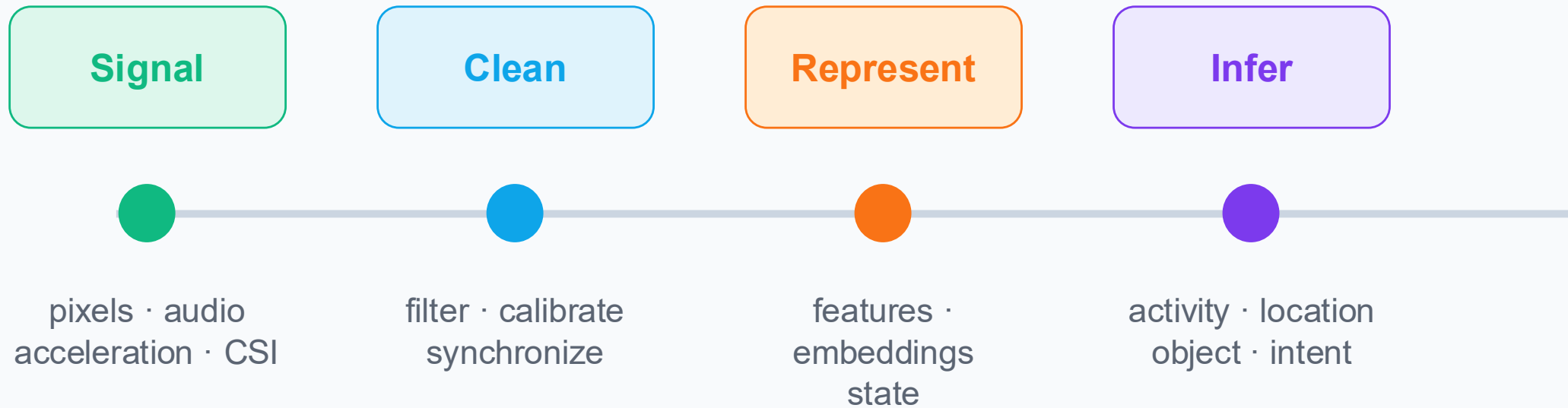
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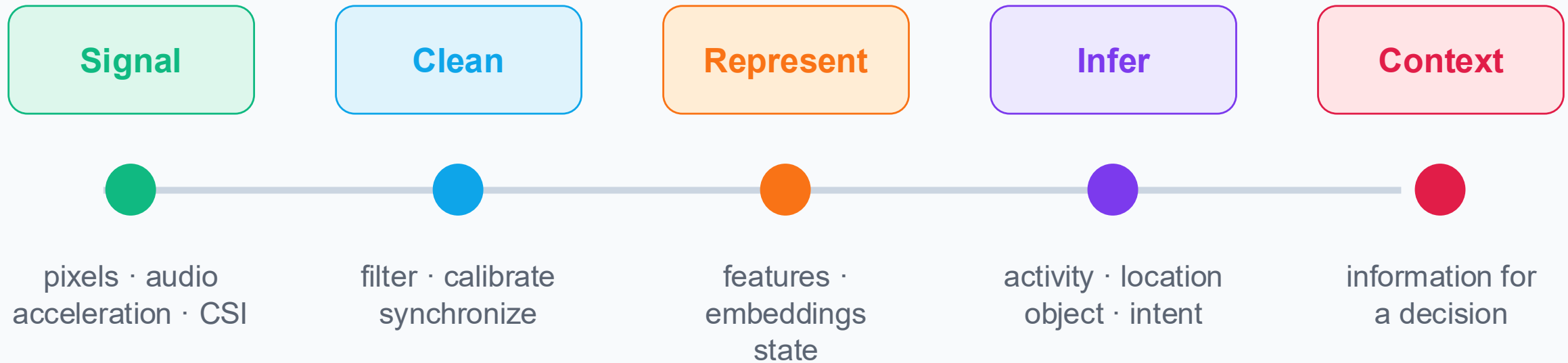
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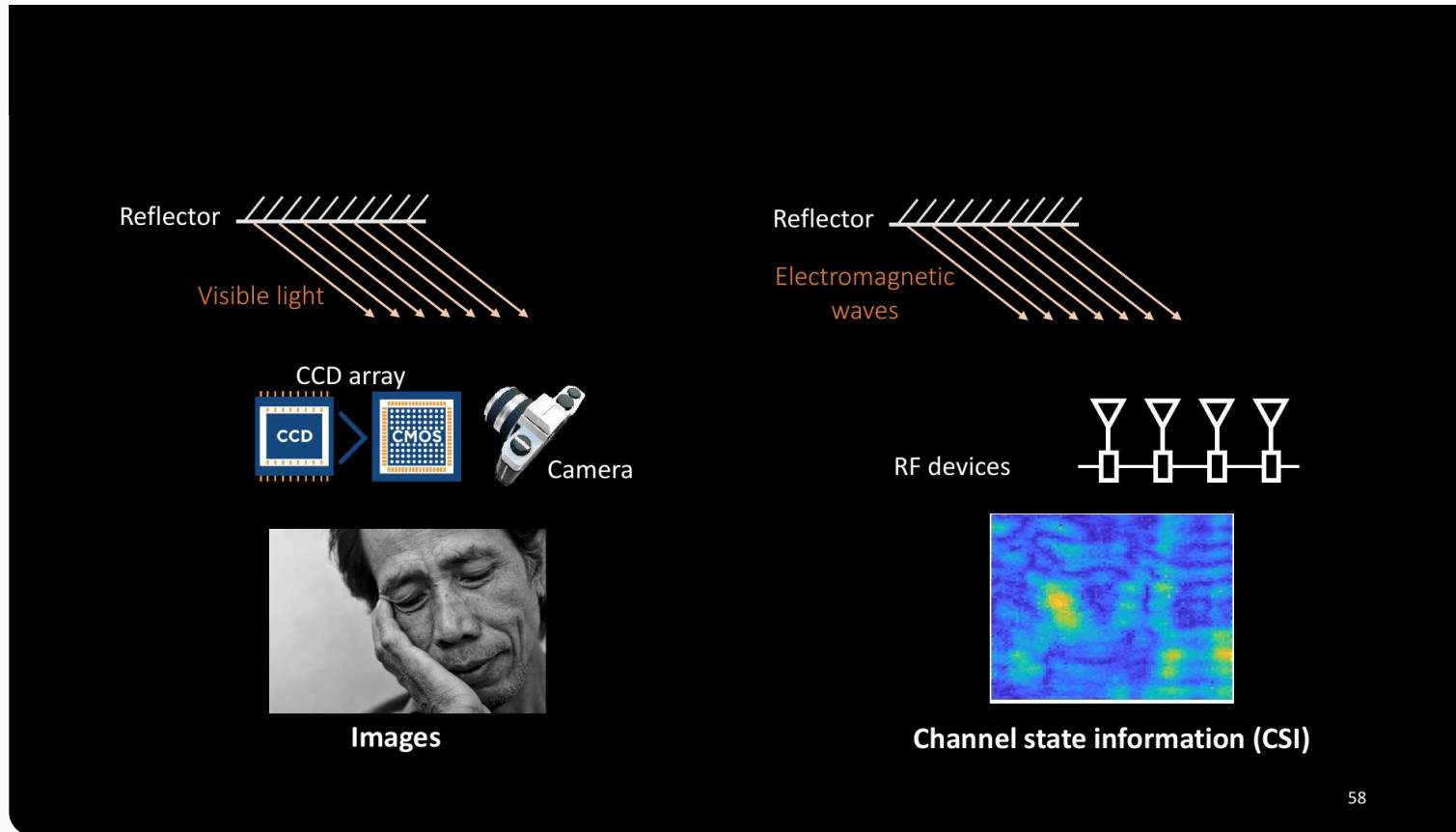


Raw signals require interpretation



Sensing accuracy depends on the complete pipeline—not only the final model.

Radio signals can communicate and sense



Communication asks:

What data was sent?

Sensing asks:

What changed the signal?

One signal can carry data and reveal context.

03

Computing coordinates resources

The mobile platform runs tasks across heterogeneous processors and across device, edge, and cloud.

Resource constraints

Runtime orchestration

Execution placement

Five constraints shape every mobile design

01



Energy

How long can it
run?

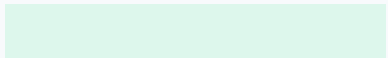
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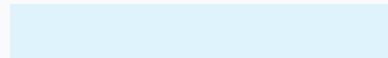


02



Latency

How soon must it
respond?



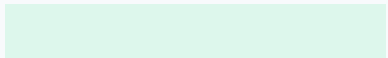
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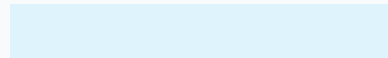


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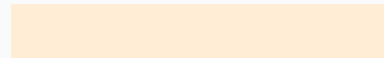


03



Privacy

What data can
leave the device?



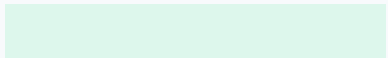
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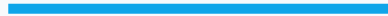


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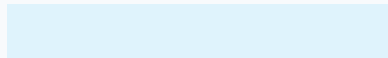


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Latency

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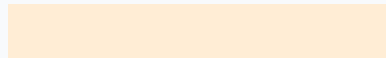


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Privacy

What data can leave the device?



04



Memory

Can data and models fit?



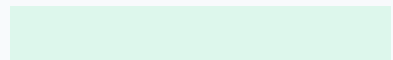
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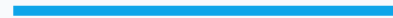


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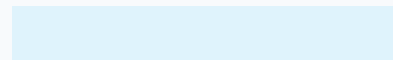


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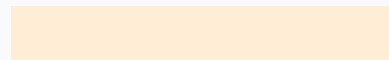


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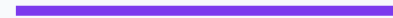


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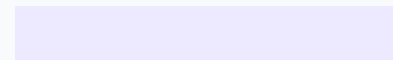


04



Memory

Can data and models fit?



05



Bandwidth

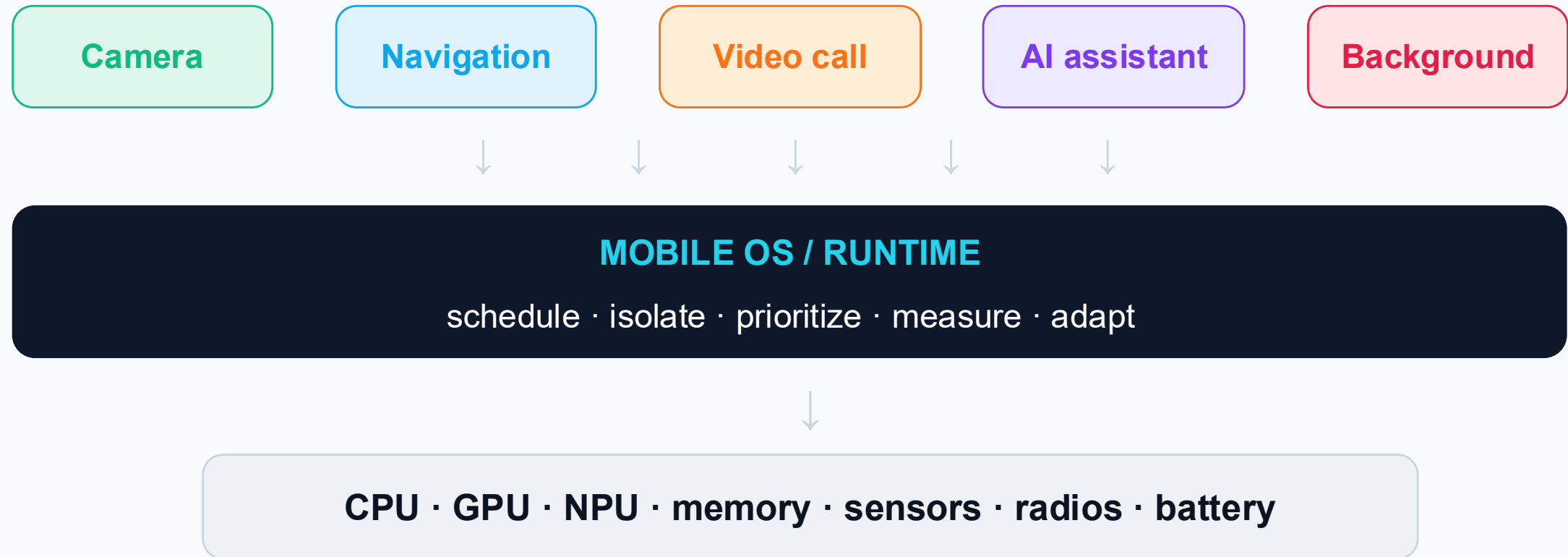
What can move over the network?



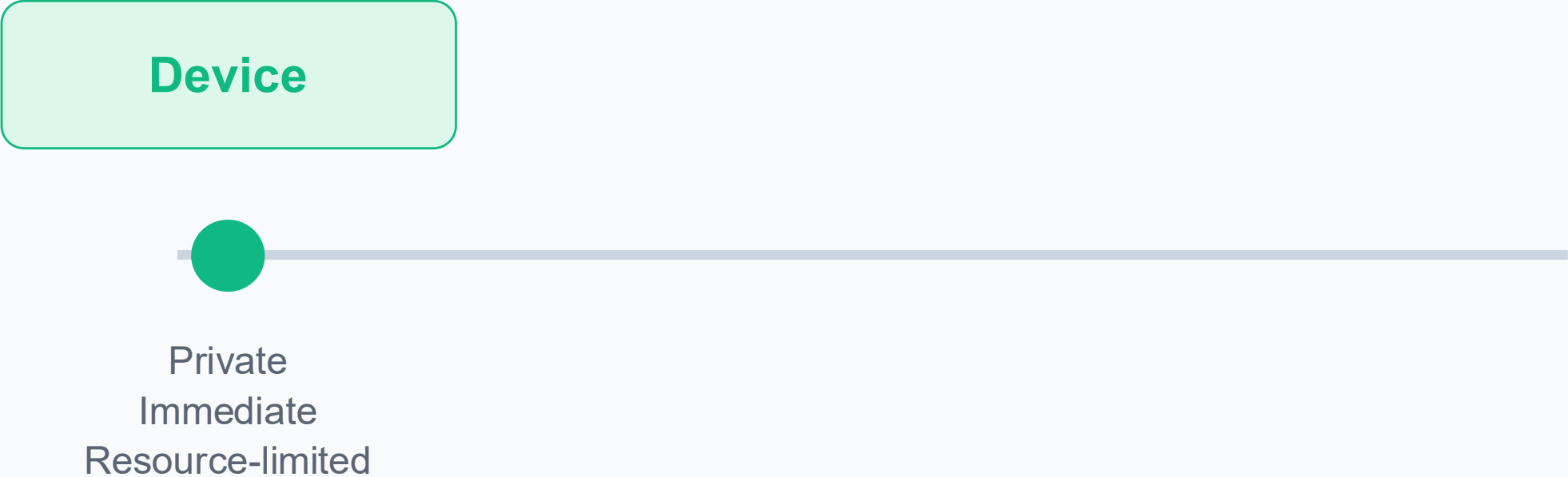
Optimizing one resource often consumes another.

The OS coordinates competing workloads

Applications



Computation spans device, edge, and cloud

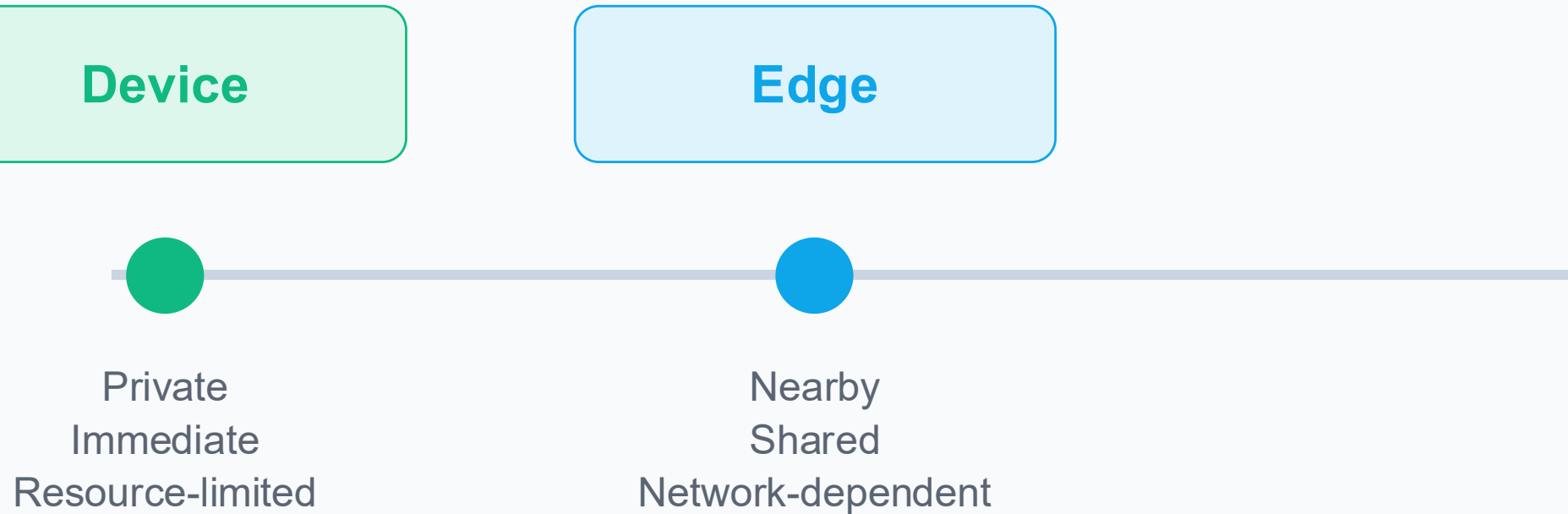


The diagram illustrates the execution continuum as a horizontal line with a green circle at the left end. Above the circle is a light green rounded rectangle labeled 'Device'. Below the circle are the labels 'Private', 'Immediate', and 'Resource-limited' stacked vertically.

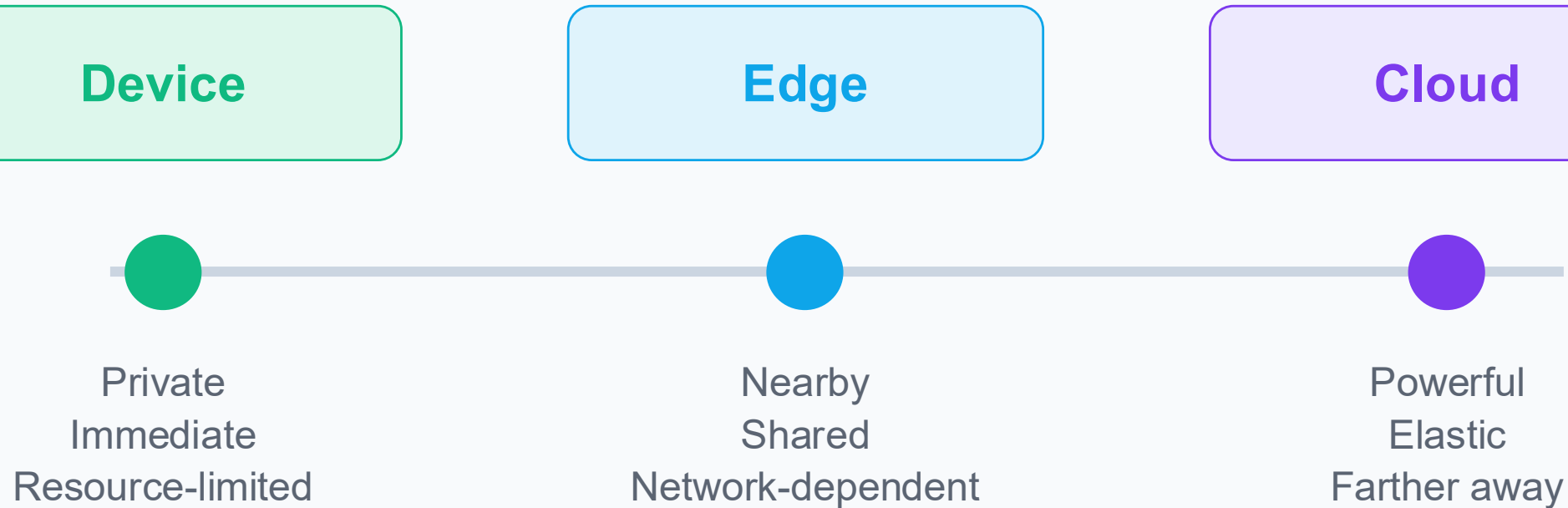
Device

Private
Immediate
Resource-limited

Computation spans device, edge, and cloud



Computation spans device, edge, and cloud



Moving computation changes latency, energy, bandwidth, and privacy at the same time.

Offloading is a decision—not a default

Example task: recognize objects from a live camera

Run on device

- No upload delay
- Raw data stays local
- Uses battery and compute budget
- Limited model size

Send to edge / cloud

- More compute and memory
- Can use a larger model
- Consumes network capacity
- Adds delay and privacy exposure

The best placement depends on the current context—not a permanent rule.

A good model can still feel bad

96%

Model accuracy

but...

1.8 s latency

The feedback arrives too late.

A good model can still feel bad

96%

Model accuracy

but...

1.8 s latency

The feedback arrives too late.

High power draw

The battery cannot sustain it.

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Model accuracy

but...

1.8 s latency

The feedback arrives too late.

High power draw

The battery cannot sustain it.

Runtime contention

Competing workloads disrupt performance.

A good model can still feel bad

96%

Model accuracy

but...

**Mobile systems are
evaluated end to end.**

1.8 s latency

The feedback arrives too late.

High power draw

The battery cannot sustain it.

Runtime contention

Competing workloads disrupt performance.

Raw-data upload

The privacy cost is unacceptable.

04

Intelligence turns context into action

Models recognize what is happening; policies decide what the system should do next.

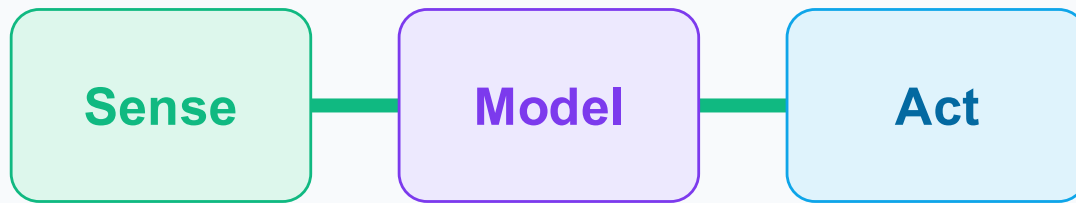
On-device inference

Efficient models

Adaptive decisions

On-device intelligence closes the loop faster

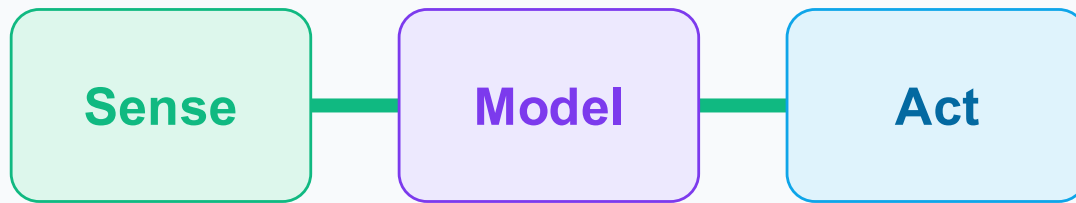
On device



Fast response · works offline · raw data stays local

On-device intelligence closes the loop faster

On device



Fast response · works offline · raw data stays local

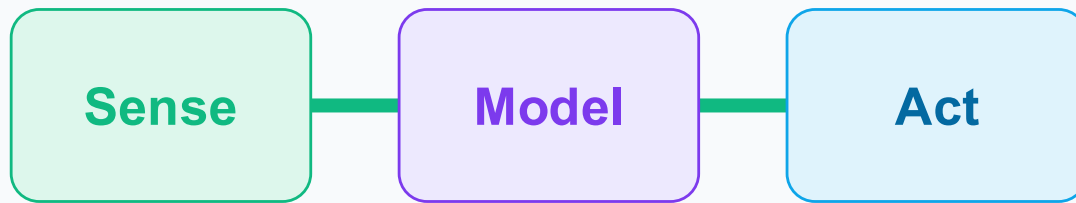
Cloud-dependent



More compute · larger models · added delay and dependency

On-device intelligence closes the loop faster

On device



Fast response · works offline · raw data stays local

Cloud-dependent



More compute · larger models · added delay and dependency

Local and remote intelligence are complementary design choices.

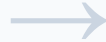
Large models must fit small devices

Large model



Large models must fit small devices

Large model



Quantize
Prune
Distill
Compress

Large models must fit small devices

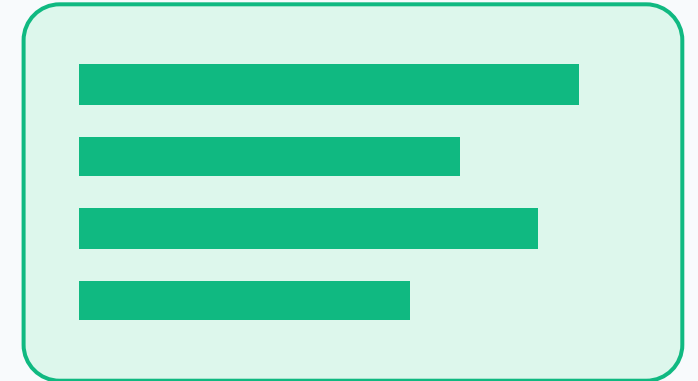
Large model



Quantize
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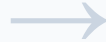


Mobile model

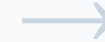


Large models must fit small devices

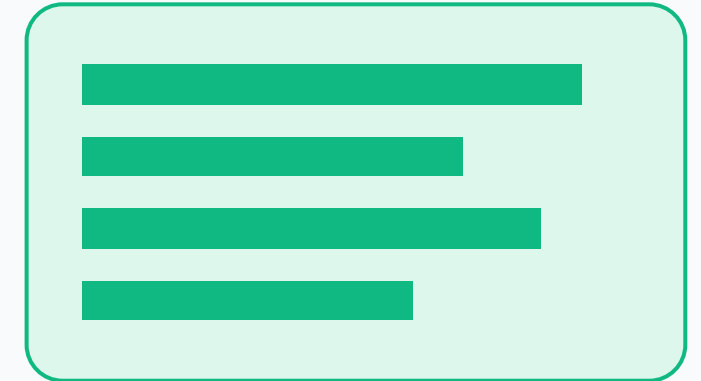
Large model



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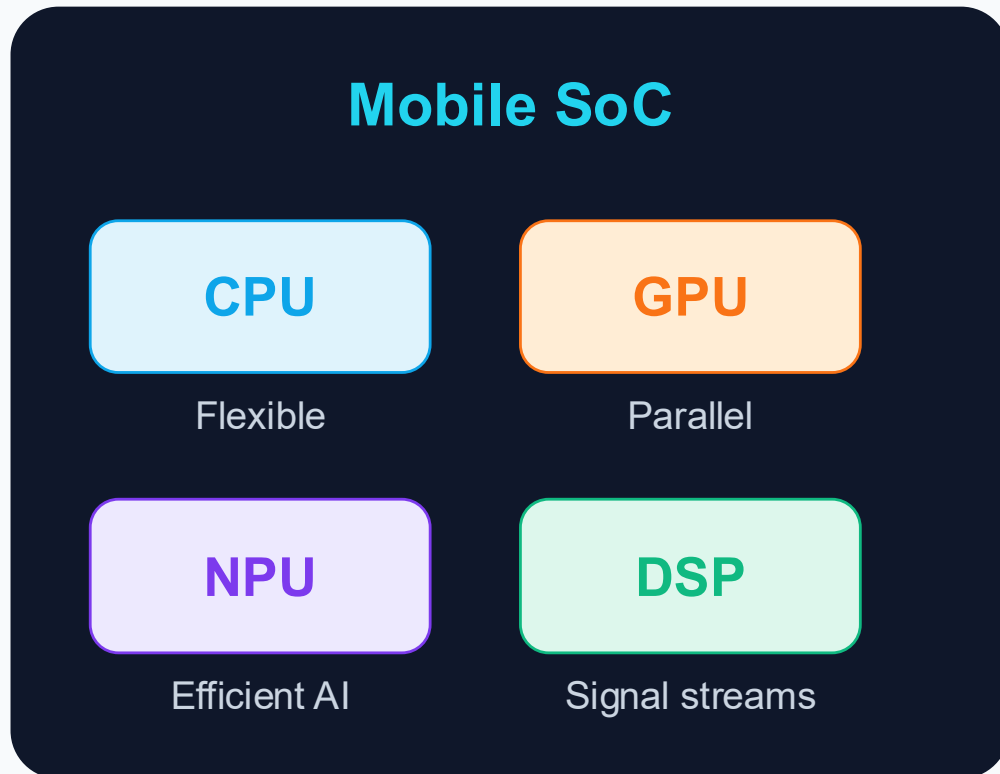


Mobile model



The goal is not the smallest model—it is the best end-to-end trade-off.

Hardware shapes model design



Model choices must match:

Supported operations
Memory movement
Parallelism and batch size
Precision and quantization
Power and energy budget

Hardware–software co-design is a mobile systems problem.

Mobile intelligence should adapt at runtime

Observe

**Battery · network · location ·
workload**

Mobile intelligence should adapt at runtime

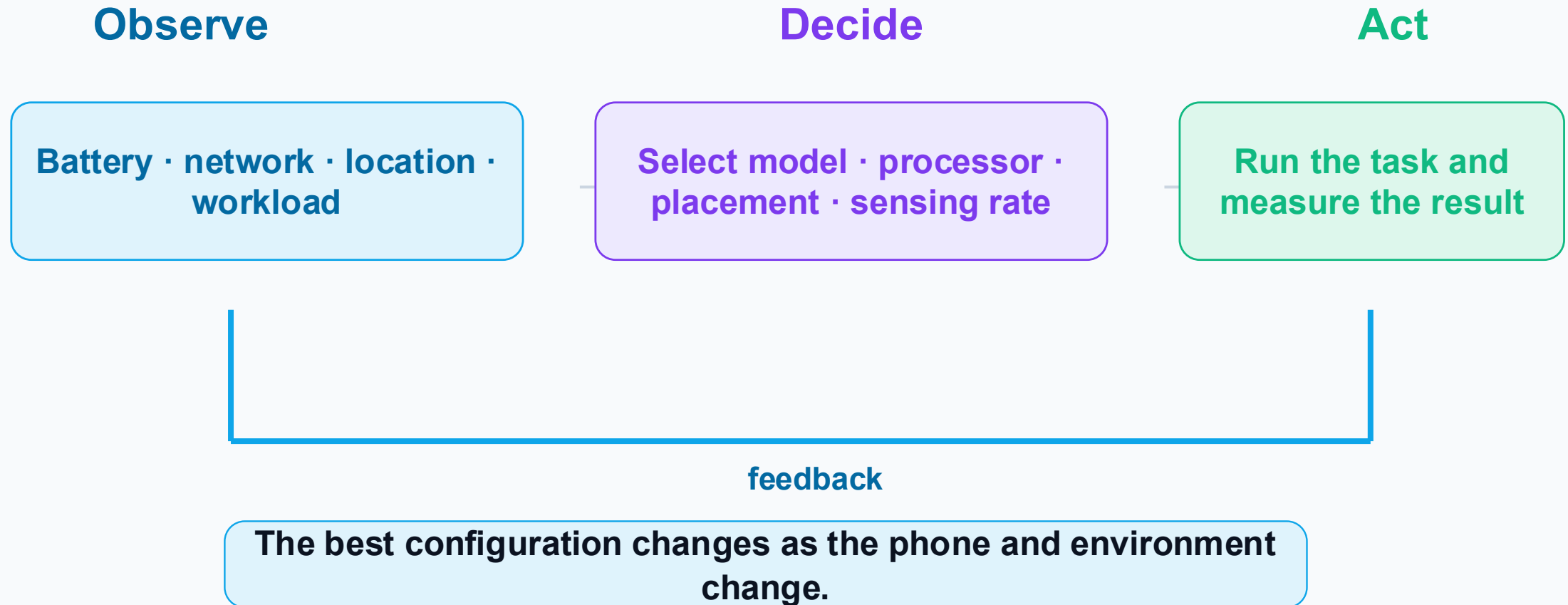
Observe

Battery · network · location ·
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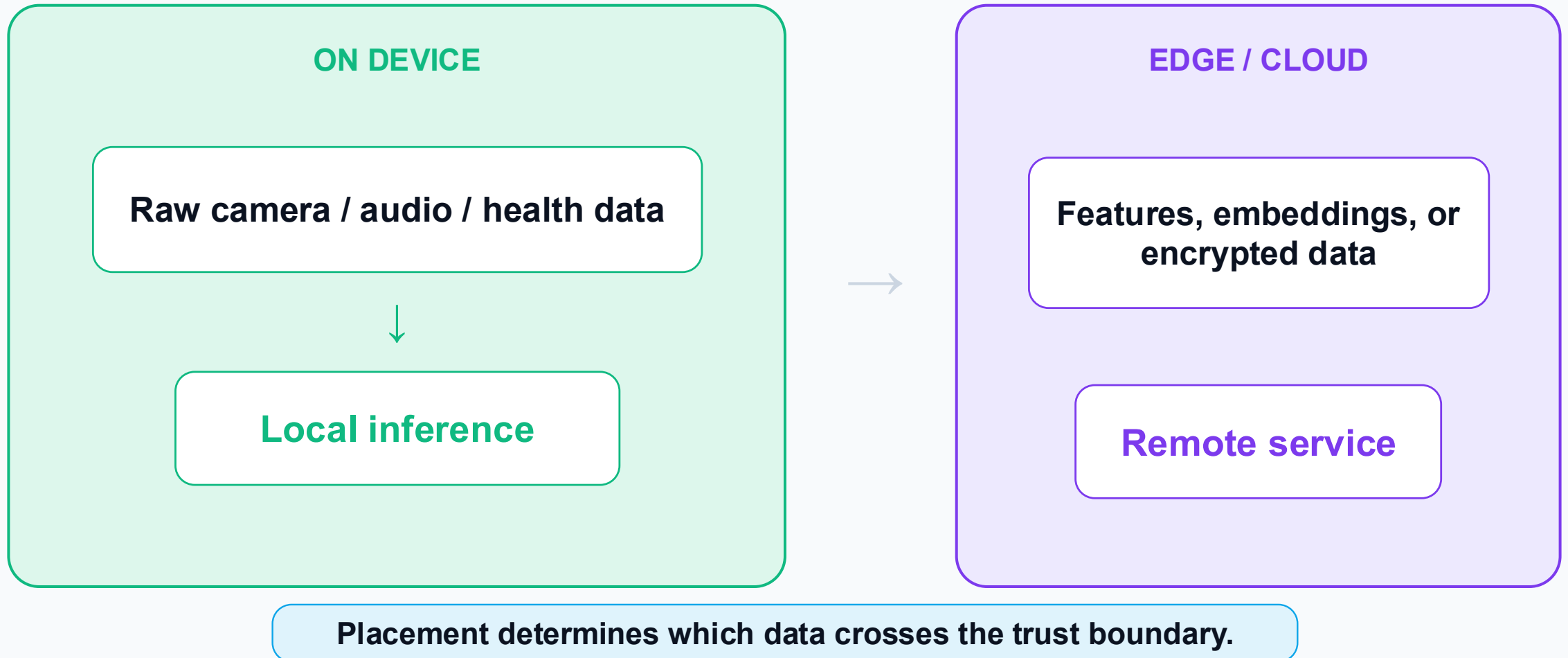
Decide

Select model · processor ·
placement · sensing rate

Mobile intelligence should adapt at runtime



Privacy changes where intelligence should run



The course follows the information-to-action loop

01



Communication

Wireless systems
MIMO / OFDM
mobile networks

The course follows the information-to-action loop

01

Communication

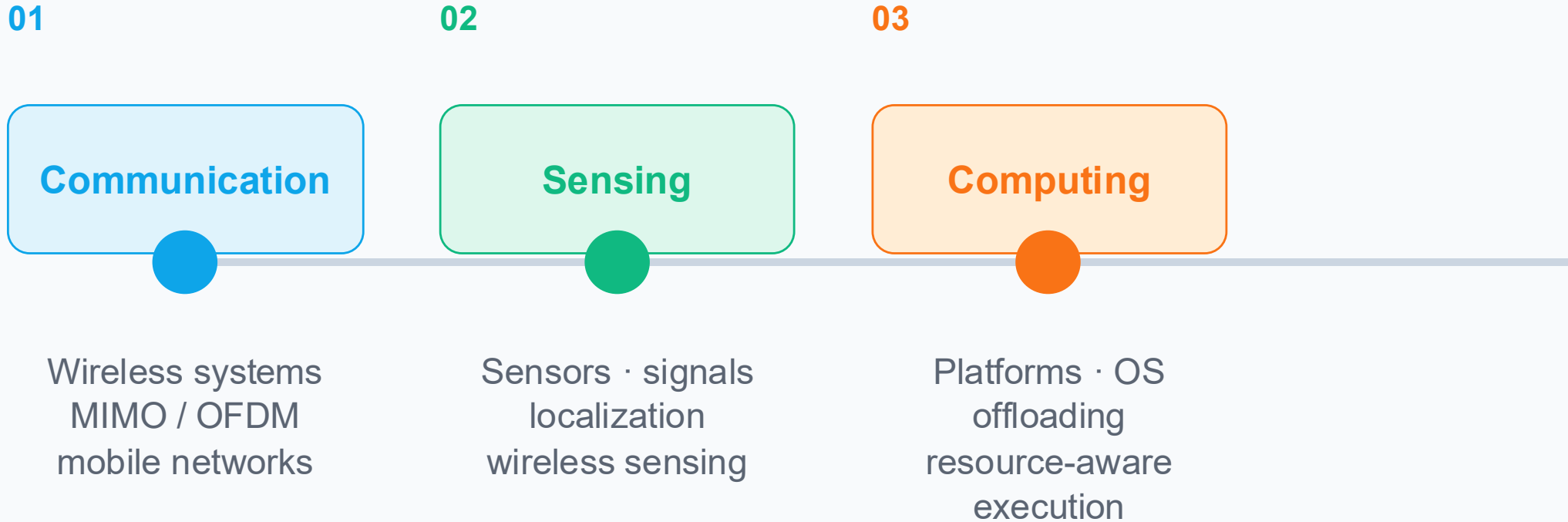
Wireless systems
MIMO / OFDM
mobile networks

02

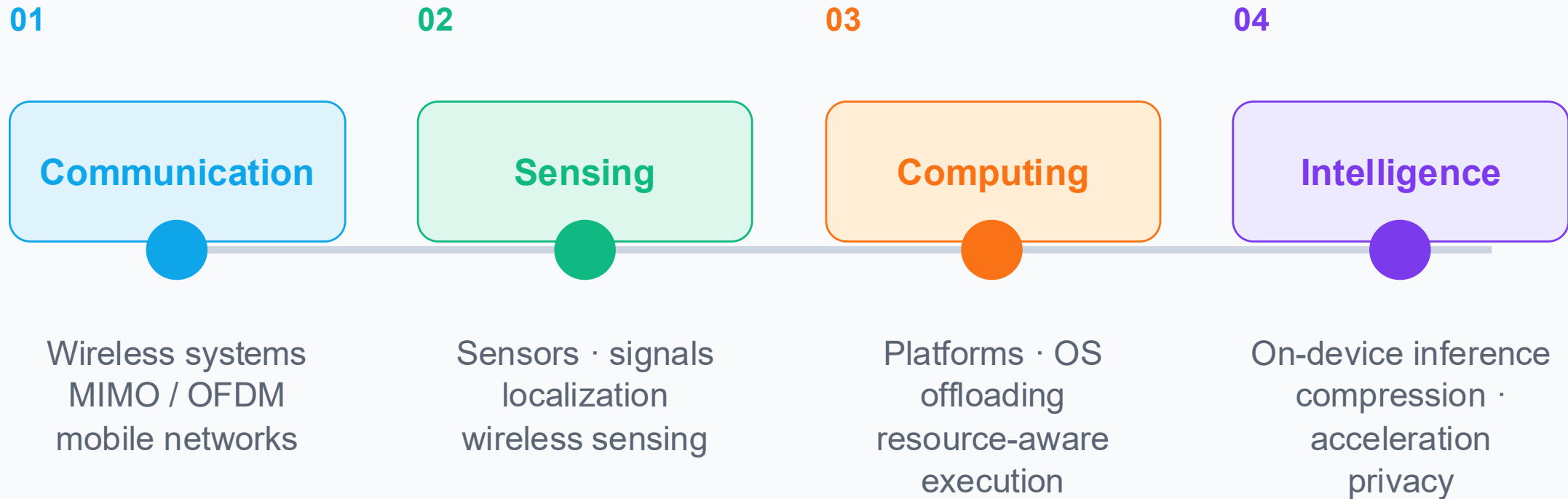
Sensing

Sensors · signals
localization
wireless sensing

The course follows the information-to-action loop



The course follows the information-to-action loop



The modules build one end-to-end mobile and edge system.

We learn through papers, systems, and evidence

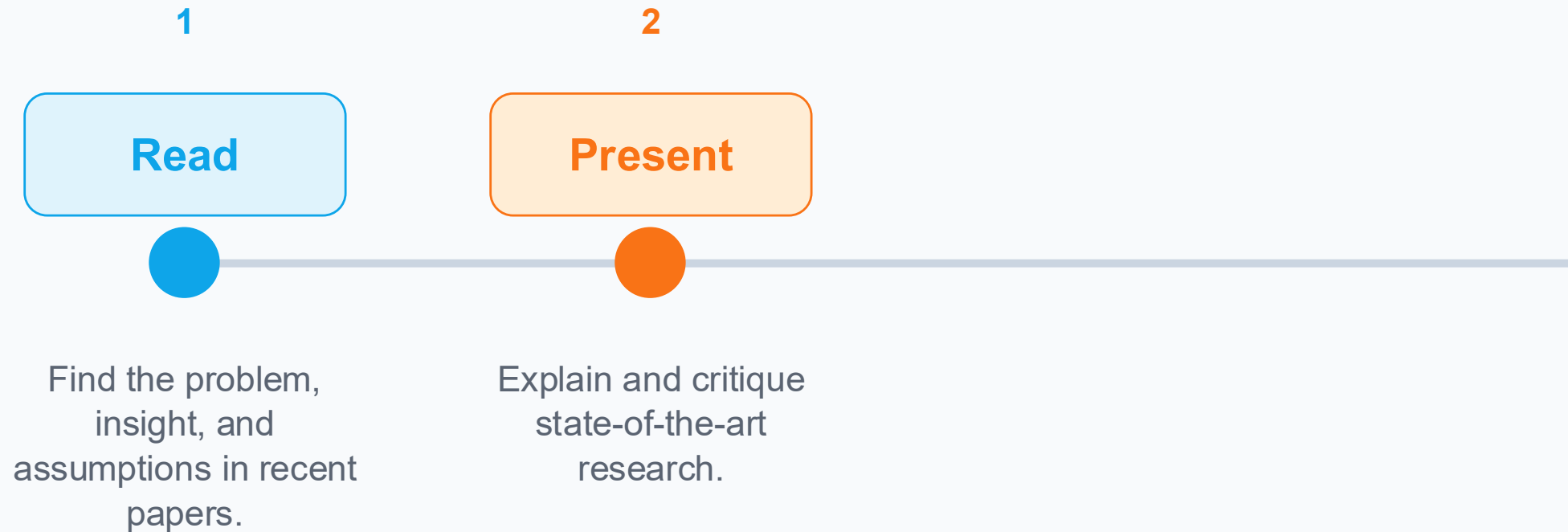
1

Read

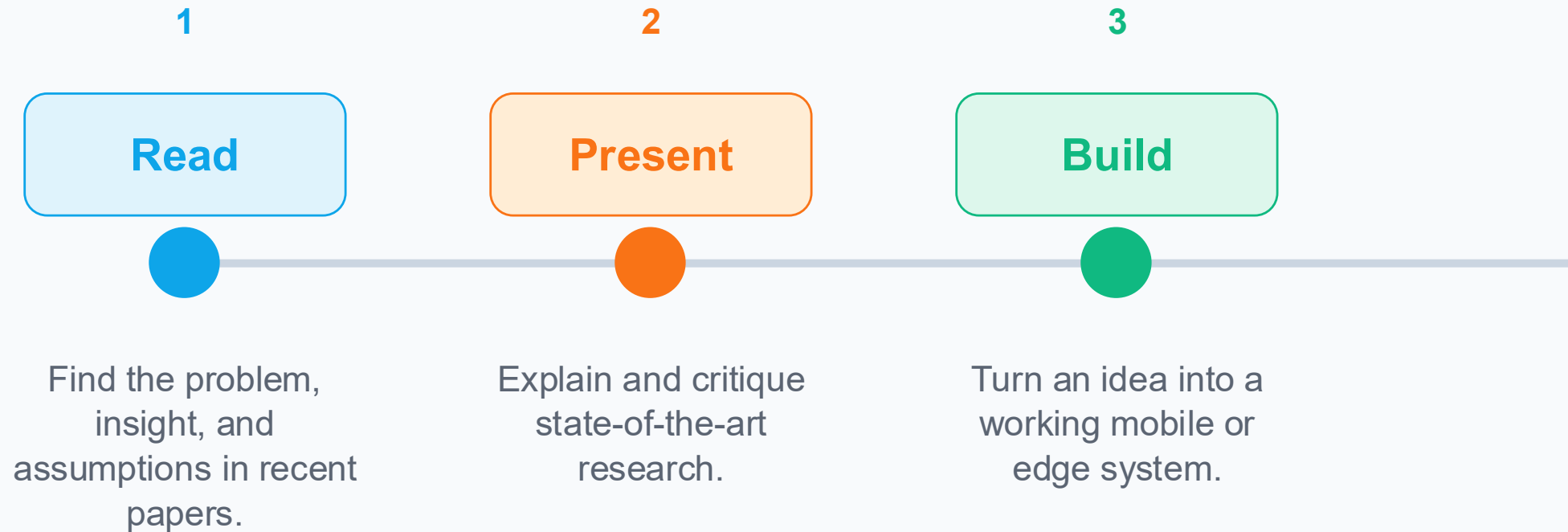


Find the problem,
insight, and
assumptions in recent
papers.

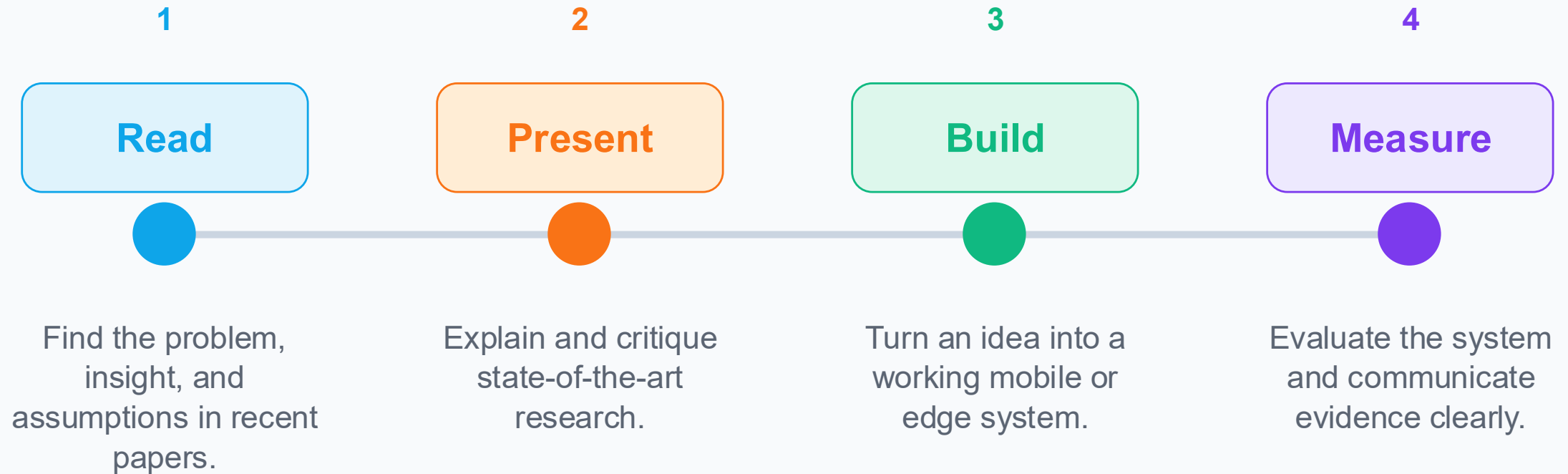
We learn through papers, systems, and evidence



We learn through papers, systems, and evidence



We learn through papers, systems, and evidence



The semester project integrates the complete research cycle.

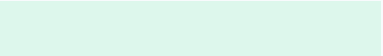
Course logistics at a glance

01



Class

Wednesday
4:00–6:50 PM



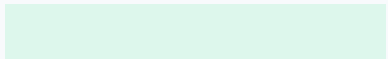
Course logistics at a glance

01

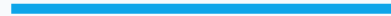


Class

Wednesday
4:00–6:50 PM

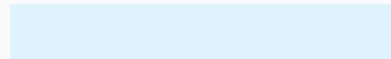


02

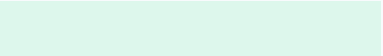
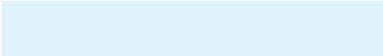
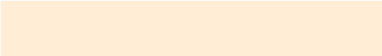


Location

Park Hall
Room 148



Course logistics at a glance

01	02	03
		
Class	Location	Piazza
Wednesday 4:00–6:50 PM	Park Hall Room 148	Access via UBLeads
		

Course logistics at a glance

01	02	03	04
Class	Location	Piazza	Background
Wednesday 4:00–6:50 PM	Park Hall Room 148	Access via UBLearns	OS · networks basic ML

Course logistics at a glance

01

02

03

04

05

Class

Wednesday
4:00–6:50 PM

Location

Park Hall
Room 148

Piazza

Access via
UBLearns

Background

OS · networks
basic ML

Coding

Python and/or
C/C++

Office hours: by appointment · Davis Hall 321

Grading rewards research and system building

1

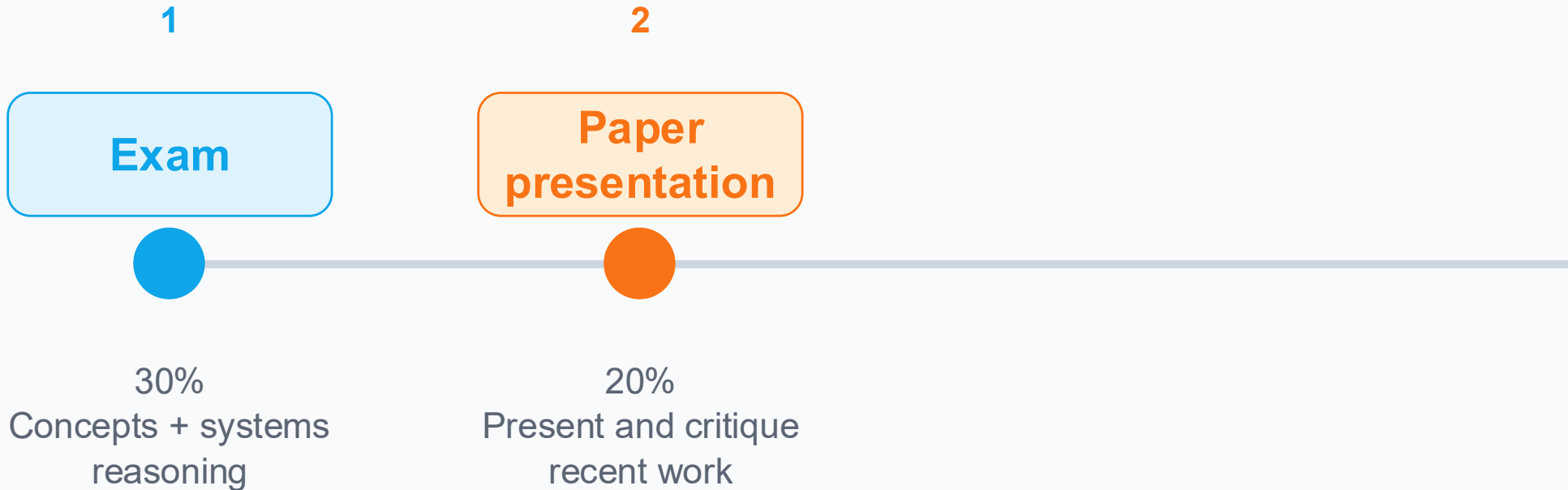
Exam



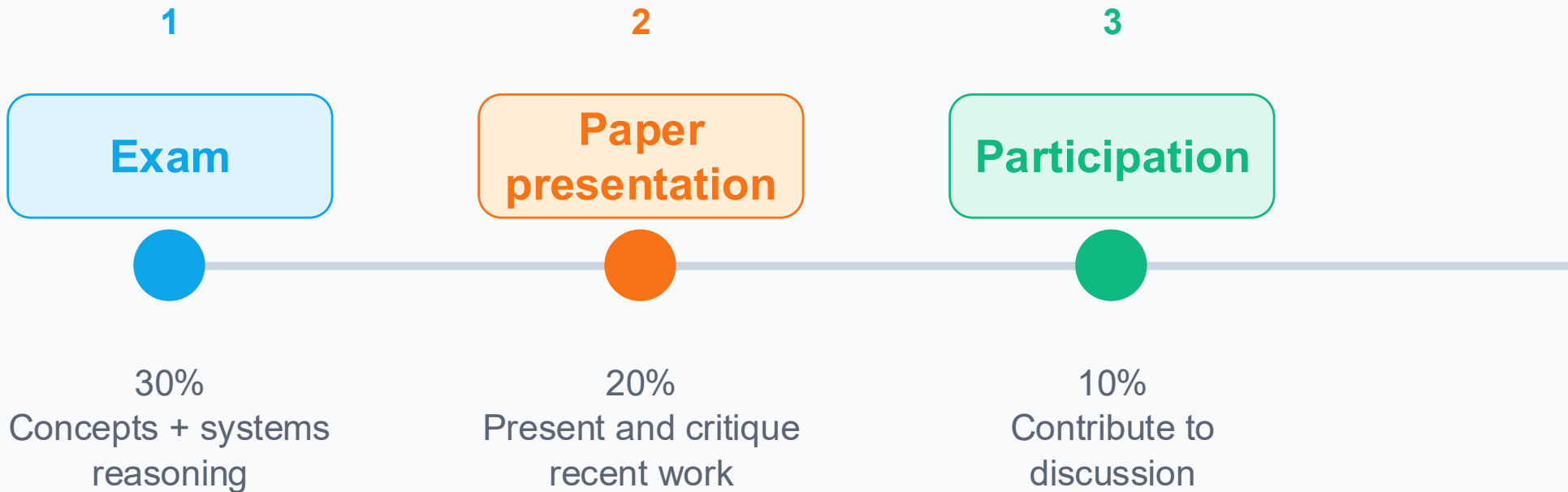
30%

Concepts + systems
reasoning

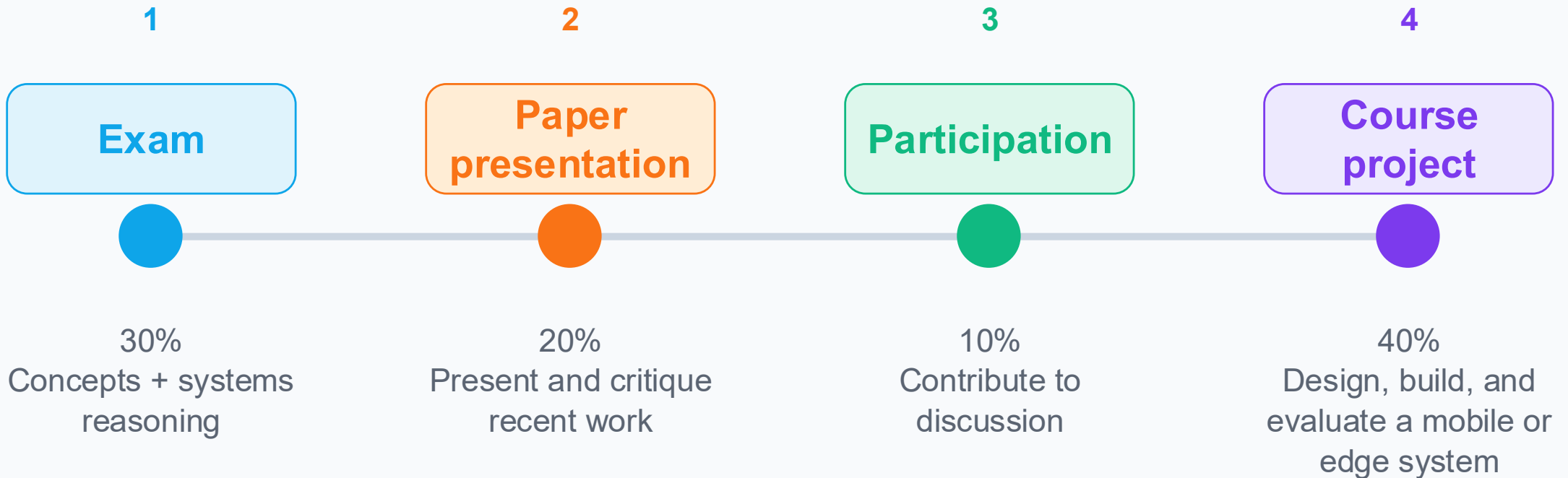
Grading rewards research and system building



Grading rewards research and system building



Grading rewards research and system building



The course project is the largest component.

Mobile systems turn change into reliable action

- 01 — Communicate — reach beyond the device
- 02 — Sense — understand the local world
- 03 — Compute — work within resource limits
- 04 — Decide → execute — act and adapt

Build the system—not just the component.

